

Sound Attenuators

Sound attenuators are used to reduce the noise generated by fans and other equipment. They can be used in a variety of applications, including HVAC systems, industrial facilities, and commercial buildings.

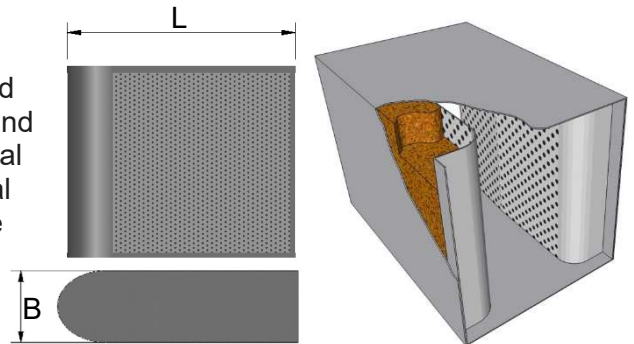


Construction:

Sound attenuators typically consist of two main components: absorption modules and a casing. The absorption modules are made of a material that is good at absorbing sound, such as glass fiber or mineral wool. The casing is made of a material that is strong and durable, such as galvanized iron.

Absorption modules (Splitters):

Absorption modules are the most important part of a sound attenuator. They are the ones that actually absorb the sound waves. Absorption modules are typically made of a material that has a high surface area, such as glass fiber or mineral wool. The high surface area allows the sound waves to be scattered and absorbed.

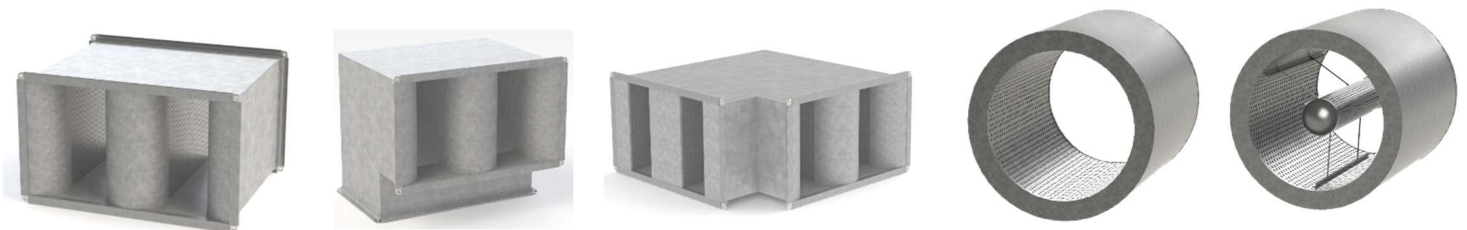


Casing:

The casing is the outer shell of a sound attenuator. It protects the absorption modules and helps to keep the sound from escaping. The casing is typically made of a strong and durable material, such as galvanized iron.

Shapes:

Sound attenuators can be made in a variety of shapes. The most common shape is rectangular. Other shapes include circular, oval, and angled. The shape of the sound attenuator will affect its performance. For example, a rectangular sound attenuator will be more effective at reducing noise than a circular sound attenuator.

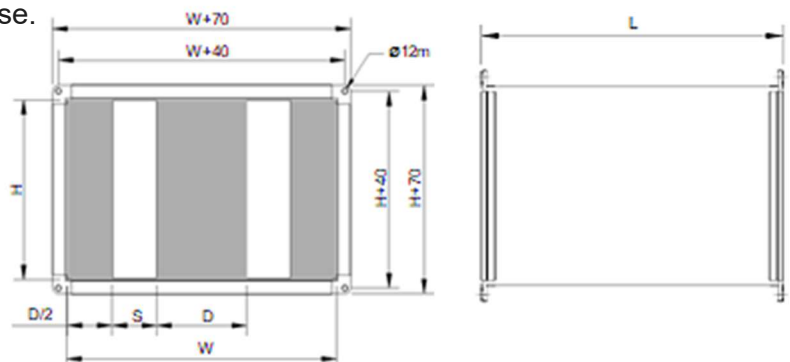


Dimensions:

The dimensions of a sound attenuator will affect its performance. The larger the sound attenuator, the more effective it will be at reducing noise.

Sound Attenuator Rectangular RSAR

- $50\text{mm} \leq S \leq 200\text{mm}$
- $100\text{mm} \leq D \leq 200\text{mm}$



Performance:

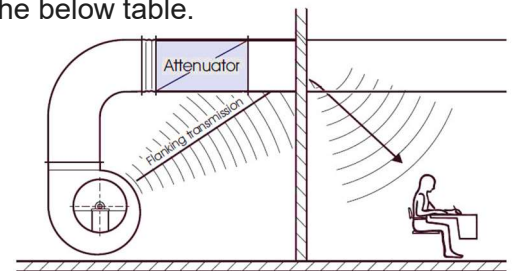
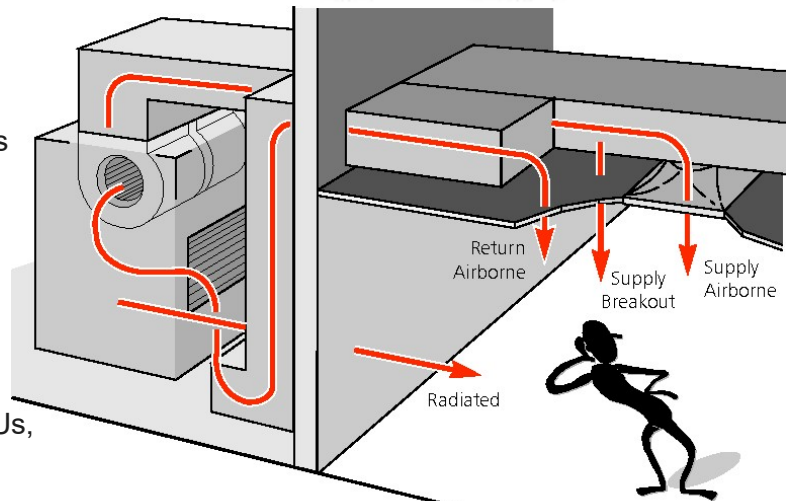
In a HVAC system, the potential noise sources are AHU, terminal units and room air outlets. The sound wave generated by these sources transfer its energy throughout the duct and to the occupant's area as shown.

Sound Power Output

In HVAC systems, elements like AHUs, PACUs, fans, and generators are noise culprits.

Sound power level (SWL): the raw noise output, independent of surroundings. AHU equipment supplier provides sound power data as given in the below table.

	63Hz	125Hz	250Hz	500Hz	1kHz	2kHz	4kHz	8kHz
Inlet + Casing	84	84	76	75	63	66	56	47
Discharge	98	91	88	91	87	86	78	72
Casing	83	83	71	74	61	59	54	36



Frequencies:

Spectrum: Noise isn't a single sound but a combination of multiple frequencies. Typically, AHUs are measured across a range of frequencies, often in octave bands.

Octave Bands: These divide the audible range (20 Hz to 20 kHz) into logarithmically spaced intervals, with each band encompassing one octave (doubling in frequency). Common octave bands for AHUs include 63 Hz, 125 Hz, 250 Hz, 500 Hz, 1000 Hz, 2000 Hz, 4000 Hz, and 8000 Hz.

Insertion loss:

Insertion loss is the decrease in sound pressure levels that can be expected when a silencer is inserted into the path between the source and the receiver.

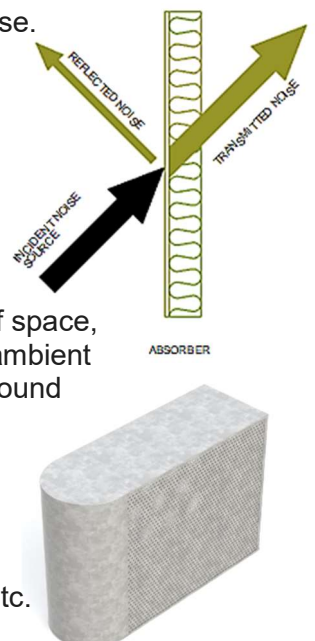
The performance of a sound attenuator is measured by its insertion loss. Insertion loss is the difference in sound pressure level before and after the sound attenuator is installed. The higher the insertion loss, the more effective the sound attenuator is at reducing noise.

Number of absorption modules:

The number of absorption modules in a sound attenuator will also affect its performance. The more absorption modules, the more effective the sound attenuator will be at reducing noise.

Sound Pressure is dependent on the acoustic environment. The factors involved include the effects of nearby reflecting surfaces, receiver distance, type of space, the amount of location in the space, the presence of barriers, and the intrusion of ambient sounds. Therefore, the sound pressure resulting from a given AHU — generated sound power depends on:

- ◆ Distance from the AHU to the room
- ◆ The size of the room
- ◆ The absorptive properties of interior furnishings
- ◆ Attenuating elements such as silencers, duct liner, duct branches, elbows, etc.



Selection:

Selecting a sound attenuator entails greater complexity compared to other products. To ensure an efficient and successful selection process, involving the Customer Support Department (CSD) is highly recommended.

However, to expedite selection and avoid potential delays, providing complete information upfront is crucial. Incomplete data necessitates repeated requests, jeopardizing project timelines and customer satisfaction.

Therefore, we encourage proactive communication and data verification prior to CSD engagement.

There are two main ways to select a sound attenuator:

- **Quick Selection (Based on independent performance):** In this case, we select a sound attenuator (no. of baffles, length) that specifies a recommended insertion loss.
 - ◆ **CSD Requirements:** RSA sizes (WxH) & Flow rate.
- **Noise Study (Based on system noise level):** In this case, the customer provides the manufacturer with information about the system. The manufacturer then (Rabie) uses this information to calculate the insertion loss required to meet the desired noise level.

CSD Requirements:

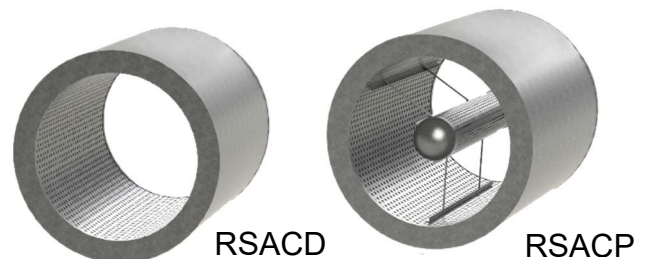
- ◆ **Data from the equipment:** The data should include the sound power levels (SWL) at each of the octave bands (Frequency), as well as the airflow rate.
- ◆ **Complete Schematic drawing layouts:** showing (Complete Architecture layout, All floors layout, Air flow at each branch, Dimensions of the air ducts, Vertical heights of the rooms and ducts, TAGs).
- ◆ **Any Special requirements needed by Customer:** (Maximum NC level, Maximum allowable pressure drop value, Maximum allowable attenuator length,.....).

Approval:

The customer must approve the sound attenuator selection before it is manufactured. This approval includes the size, performance, and pressure drop of the sound attenuator.

Round Sound Attenuators:

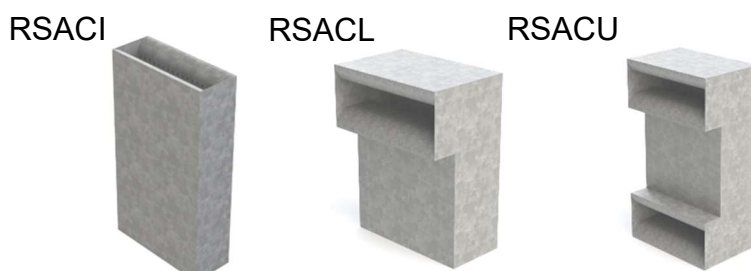
Suitable for installation on round ducts, features two designs: one with a central pod for enhanced noise reduction and another without for simpler installation.



Cross Talk Sound Attenuators:

type attenuators have been designed to avoid noise interference between interconnected rooms. Air flow is permitted whilst noise transfer is limited.

Type RSACL, RSACU, RSACI are used as air transfer attenuators between adjoining area where the acoustic integrity of a common partition or ceiling needs to be maintained.



Summary: Sound attenuators are an important part of HVAC systems and other applications where noise reduction is required. By selecting the right sound attenuator, you can ensure that your system meets your noise requirements.

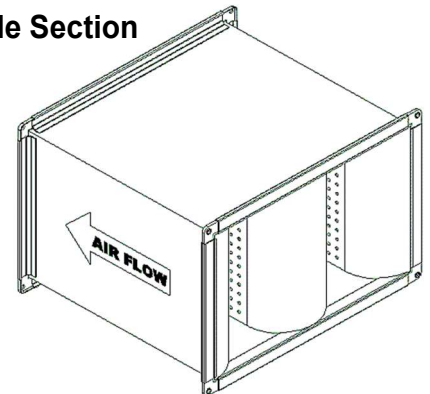
Installation:

We offer sound attenuators in both single and multi-section configurations to meet your specific noise reduction requirements.

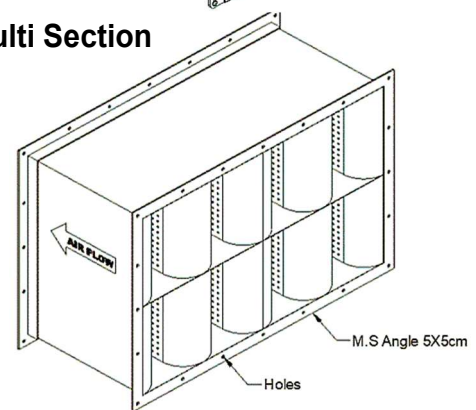
Single-Section Attenuators:

- Ideal for smaller applications with limited space.
- Maximum dimensions: 2 meters wide and 1.2 meters tall.
- Feature duct flanges on both ends for easy connection with screws and sealing tape.

Single Section



Multi Section



Multi-Section Attenuators:

- Designed for larger spaces and higher noise reduction needs.
- Exceed the size limitations of single sections (2m x 1.2m).
- Delivered as multiple units easily assembled on-site.
- Typical interconnection of units in a multiple-unit sound attenuator is using an angular welded flange of 50×5mm.

Take special care to seal between the edges of each unit to prevent air from passing between the units.